



PUSL3190 Computing Project

Project Proposal

Hybrid Hardware Store Management System

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Problem Statement

The owner of a local hardware store is currently using an outdated POS system that was once suitable for basic operations but has now become a major obstacle to efficient business management. Although the system can handle simple billing and receipt generation, it frequently crashes, produces transaction errors, and causes delays during customer checkouts. These technical issues lead to inaccurate financial records, slower service, and frustration among both employees and customers.

Inventory management has also become increasingly difficult. The existing system often fails to update stock levels correctly or generate accurate inventory reports. As a result, the store faces issues such as stock mismatches, delayed restocking, and confusion during audits. These problems not only waste valuable time but also reduce overall productivity and profitability.

Another serious limitation is the system's inability to support multiple cashiers simultaneously. During peak hours, only one cashier can operate the system, creating long queues and negatively affecting customer satisfaction. Moreover, the current POS operates in isolation; it lacks integration with other key business areas such as employee attendance, shift scheduling, and financial tracking. The owner must therefore rely on separate manual records and spreadsheets to manage daily operations, which increases the likelihood of human error and makes data management inefficient.

Adding to these challenges, the system offers no remote access or real-time data analysis. The owner cannot monitor sales, inventory, or financial performance while away from the store, forcing them to make important business decisions based on incomplete or outdated information. This lack of visibility and automation prevents the business from growing efficiently and adapting to modern retail needs.

In short, the current POS system fails to meet the operational, managerial, and analytical requirements of the hardware business. It causes delays, inaccuracies, and inefficiencies across multiple areas from billing and inventory to employee management and finance, creating a pressing need for modernization.

Project Description

The proposed project, **Hybrid Hardware Store Management System (Local-first with Cloud enabled Features)**, is aimed at developing a complete and intelligent management solution tailored for small and medium-sized hardware stores. The main goal of this system is to bring together every essential function of the business including Point of Sale (POS), inventory control, finance, employee attendance, salary processing, and data analytics under one seamless and easy to use platform.

Today, many hardware store owners still depend on outdated or incomplete systems that only perform basic billing tasks. These systems often lack advanced functionalities such as multi-cashier support, attendance tracking, and automated financial management. As a result, store owners struggle with errors, inefficiency, and poor visibility into how their business is performing. This project is designed to address those challenges by building a hybrid system one that works locally for reliability and also connects to the cloud for advanced features such as reporting, analytics, and backup.

The hybrid approach ensures that the store can continue operating even without an internet connection while still benefiting from modern, cloud-based capabilities when online. This combination of local performance and cloud flexibility makes the system both practical and future ready. It is specifically designed for the realities of Sri Lankan hardware stores, where stable internet connections are not always guaranteed, but efficiency, speed, and accuracy are essential.

Core Modules and Features

1. Point of Sale (POS) and Multi-Cashier Management

The POS module will be the heart of the system. It will support multiple cashiers working simultaneously at different counters, each with a secure login and unique role-based permissions. The system will handle all billing operations, generate receipts, and automatically record transactions in real-time. By supporting barcode scanning and cash-based sales, the system ensures a quick and accurate checkout process. This will eliminate long queues during busy hours and help maintain customer satisfaction. Every sale will also automatically update the inventory, keeping all data synchronized between cashiers and the store's database.

2. Inventory Management

The inventory module will give the store owner complete control over stock movement. Every sale, purchase, or return will be reflected instantly in the system. Low-stock alerts will notify the owner before items run out, and overstock warnings will help avoid unnecessary expenses. The module will also generate detailed inventory reports, showing which products sell the most and which are slow-moving. Cloud synchronization will back up inventory data automatically and provide access to inventory reports remotely. This feature will allow the owner to make timely restocking decisions and maintain an optimal balance between supply and demand.

3. Finance Management

The finance module will handle day-to-day financial activities such as tracking expenses, monitoring income, and generating profit or loss statements. All financial data will be stored securely on the local database but can also be backed up to the cloud for remote access. The system will provide a clear view of daily cash flow, helping the owner understand where money is spent and how much profit the business is making. Automated financial reporting will reduce manual errors and save time that would otherwise be spent preparing spreadsheets and calculations.

4. Employee Attendance and Payroll

Managing employee attendance and salaries manually is both time-consuming and error-prone. The attendance and payroll module will automate these processes. Attendance can be marked either manually through the system or automatically using a biometric device. The payroll feature will calculate salaries based on attendance records, including allowances, deductions, and overtime. The system will also generate salary slips automatically, making payroll management more transparent and efficient. This module will help maintain fair, accurate records of employee work hours and payments while reducing administrative workload.

5. Analytics and Business Insights

One of the most powerful features of this system is its analytics module. The cloud-enabled analytics component will transform raw business data into meaningful insights. Using machine learning techniques, the system will analyze purchase patterns, identify popular product combinations, and provide recommendations for stock planning. Features such as market basket analysis will help discover products often bought together, enabling smarter bundling and promotions. Sales forecasting will use past data to predict future demand, helping the owner prepare for seasonal trends and maximize sales.

Project Objectives and Explanations

- **Enhance Reliability:**
Create an offline-first POS system that operates without internet dependency, ensuring uninterrupted billing and sales processing even during network outages.
- **Automate Key Operations:**
Streamline repetitive business tasks such as billing, payroll, and inventory tracking through automation, significantly reducing manual workload and errors.
- **Enable Smart Decision-Making:**
Integrate analytical dashboards that provide real-time visualizations of sales, profits, and stock movement to support better business decisions.

- **Support Multiple Users:**
Introduce multi-cashier functionality and role-based access control, improving team collaboration while maintaining data security.
- **Improve Financial Transparency:**
Generate automated financial reports and summaries that help the owner monitor cash flow, profitability, and expenses with accuracy.
- **Promote Business Scalability:**
Build the system architecture in a modular way so it can easily expand to manage multiple branches, products, and employees in the future.

Each of these objectives supports the vision of creating a practical, intelligent, and scalable solution that fits the modern needs of local retail environments.

Project Keywords

1. **Hybrid Systems** – Combining offline reliability with cloud-based accessibility and analytics.
2. **POS Systems** – Fast, secure, and multi-user billing solutions for efficient sales management.
3. **Inventory Management** – Automated, real-time tracking of stock movement and restocking alerts.
4. **Payroll Automation** – Streamlined employee attendance and salary calculation with minimal human intervention.
5. **Retail Analytics** – Data-driven insights for forecasting trends, understanding customer behavior, and improving business strategies.

Research Gap

Most existing POS systems used by hardware stores are either fully online or completely offline, each with major drawbacks. Cloud-based systems require a constant internet connection, while offline systems lack remote access and real-time reporting. Furthermore, current solutions focus mainly on billing and stock control but do not integrate essential features such as attendance, payroll, finance, and analytics. As a result, store owners rely on multiple tools, causing inefficiency and data inconsistencies.

High-end ERP systems offer advanced analytics but are too costly and complex for small businesses. Additionally, most software is built for general retail, not for the specific needs of hardware stores that handle diverse products and units. Therefore, there is a clear gap for an affordable, hybrid system that works both offline and online while combining POS, inventory, finance, employee management, and analytics in one unified platform.

Requirements Analysis

Technical Knowledge and Tools:

- **Frontend:** React.js or Flutter for building a responsive desktop interface.
- **Backend:** Laravel (PHP) or Node.js for handling business logic and APIs.
- **Database:** MySQL or SQLite for local storage; PostgreSQL or Supabase for cloud synchronization.
- **Cloud Integration:** Firebase, AWS, or Azure for authentication, analytics, and backups.
- **Data Analytics:** Python (pandas, scikit-learn) for sales forecasting and trend analysis.
- **Deployment:** Docker for containerization and consistent system deployment.

Hardware Devices:

- Desktop or laptop computer for development.
- Barcode scanner and receipt printer for POS testing.
- Optional biometric device for attendance tracking.

Knowledge to Learn:

- Hybrid system design and cloud synchronization.
- Role-based access control (multi-cashier support).
- Database normalization and API integration.
- Data visualization and analytics for business insights.
- UI/UX design for efficient retail interfaces.

Finance

Budget Plan (Non-reimbursable)

The project will primarily use open-source technologies, minimizing costs. However, some hardware and cloud services are required for development and testing.

Item	Description	Estimated Cost (LKR)
Development Computer	Existing personal laptop for coding and testing	–
Barcode Scanner & Receipt Printer	Required for POS testing	40,000 – 60,000
Optional Biometric Device	For attendance integration	10,000 – 15,000
Domain & SSL Certificate	For secure cloud communication	5,000 – 10,000 per year
Cloud Hosting (AWS / Firebase / Supabase)	For backups, analytics, and remote access	15,000 – 25,000 per month
Miscellaneous Expenses	Electricity, documentation, etc.	5,000 – 10,000

Total Estimated Cost: Approximately **LKR 80,000 – 120,000**

(All costs are estimates and will not be reimbursed.)

External Organizations

External Parties

- **Hardware Store Owners:**
Will be involved in the requirement-gathering and testing stages to ensure that the system meets real business needs.
- **Supervisors / Academic Mentors:**
Will provide technical guidance and feedback during each development phase.

Other Staff

- **Store Cashiers:**
Will assist in testing the POS and multi-cashier modules.
- **Accountant or Store Clerk:**
Will help test the finance and reporting modules for accuracy.

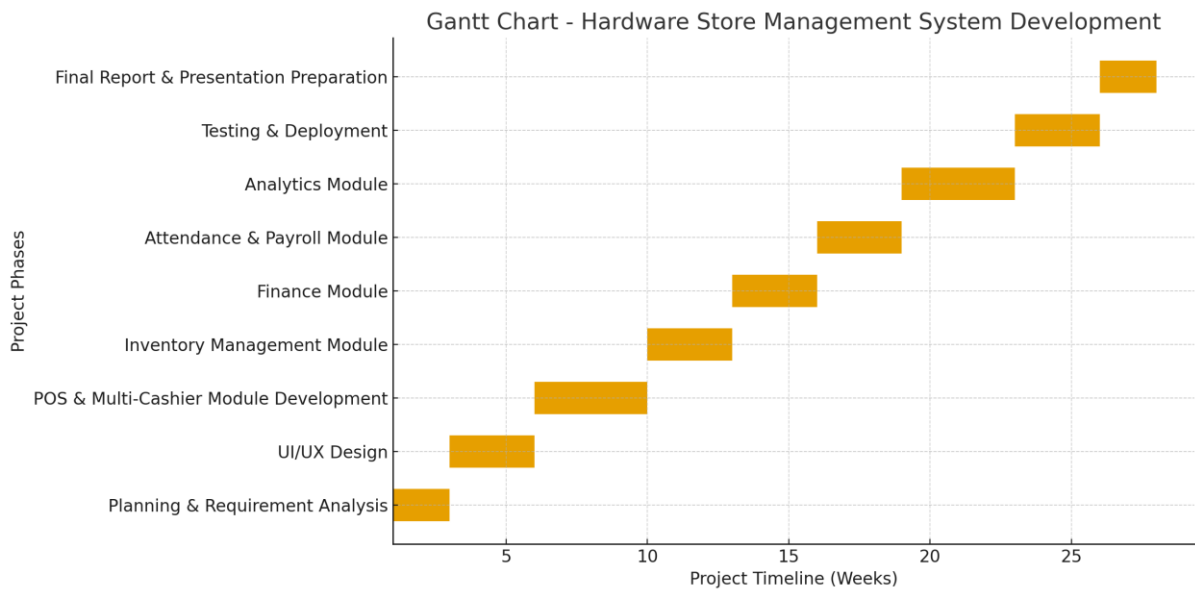
Supporting Staff

- **UI/UX Designers (if available):**
To refine the system's interface for usability and visual clarity.
- **Peers or Developers:**
To provide code reviews and assist with testing, debugging, and deployment.

Time Frame / Timeline

Phase	Duration	Expected Deliverables
Planning & Requirement Analysis	2 Weeks	Requirement documentation and system architecture design
UI/UX Design	3 Weeks	Wireframes, design mockups, and flow diagrams
POS & Multi-Cashier Module Development	4 Weeks	Core billing system, multi-cashier functionality, and receipt generation
Inventory Management Module	3 Weeks	Product tracking, stock updates, and low-stock alerts
Finance Module	3 Weeks	Expense tracking and profit/loss reporting
Attendance & Payroll Module	3 Weeks	Employee attendance, salary calculations, and payroll automation
Analytics Module	4 Weeks	Data visualization, sales insights, and demand forecasting
Testing & Deployment	3 Weeks	Debugging, documentation, and hybrid deployment setup
Final Report & Presentation Preparation	2 Weeks	Final documentation and project submission

Total Estimated Duration: ~23 Weeks (6 Months)



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